

Cambridge Lower Secondary Checkpoint

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SCIENCE

Paper 1

0893/01

April 2025

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

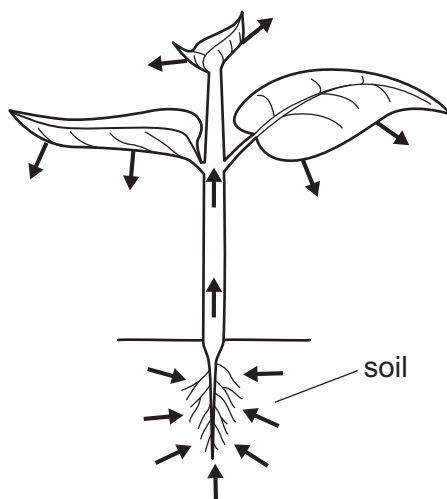
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

- 1 The diagram shows the pathway of water moving through a plant.



- (a) Describe how water enters the plant.

.....

.....

..... [2]

- (b) Write down the name of **one other** substance that enters the plant with the water.

..... [1]

- (c) Complete the sentence.

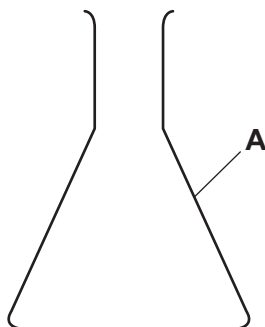
Water is lost from a plant through organs called

This process is called

[1]

- 2 Oliver reacts powdered calcium carbonate with dilute acid.

The diagram shows one of the pieces of equipment Oliver uses.



- (a) Name equipment **A**.

..... [1]

- (b) Oliver records the mass of the reactants before putting them into equipment **A**.

reactant	mass in g
powdered calcium carbonate	3
dilute acid	100

- (i) The total mass of the reactants is the **same** as the total mass of the products.

Explain why.

..... [1]

- (ii) The mass of equipment **A** and the reactants is **more than** the mass of equipment **A** and its contents after the reaction.

Explain why.

..... [1]

- (c) Oliver repeats the experiment using large pieces of calcium carbonate and **not** powdered calcium carbonate.

Write down how the rate of reaction is different using the large pieces of calcium carbonate.

..... [1]

- 3 Mike puts water on his hand.

The water evaporates and his hand feels cold.

Explain this cooling effect caused by evaporation.

Use ideas about particles.

.....

.....

.....

.....

..... [3]

- 4 This question is about ionic bonding.

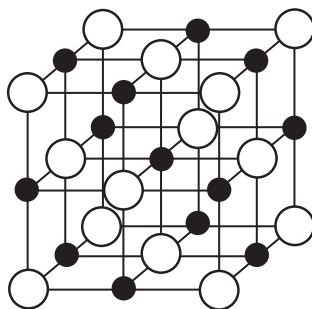
Magnesium oxide has ionic bonding between a magnesium ion and an oxide ion.

Look at the model of the structure of magnesium oxide.

Key

○ oxide ion, O^{2-}

● magnesium ion, Mg^{2+}



The structure of magnesium oxide has ionic bonds.

- (a) (i) Describe what is meant by an ionic bond.

.....
 [1]

- (ii) Describe how a magnesium atom makes a magnesium ion, Mg^{2+} , and how an oxygen atom makes an oxide ion, O^{2-} .

how a magnesium atom makes a magnesium ion

.....

how an oxygen atom makes an oxide ion

.....

[2]

- (b) An ionic bond is **not** made between two iodide ions, I^- .

Explain why.

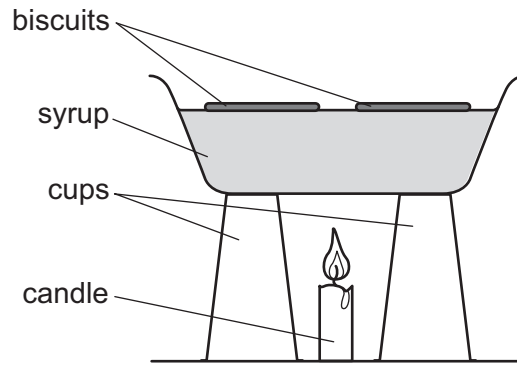
.....
 [1]

5 This question is about tectonic plates.

Oliver makes a model to demonstrate what happens at the boundary of tectonic plates.

Oliver:

- pours cold syrup into a large container
- places two biscuits on top of the syrup
- assembles the equipment as shown.



(a) Draw a straight line to match each **part of the model** to the **part of the Earth** it represents.

part of the model

biscuit

syrup

candle

part of the Earth

mantle

core

crust

[2]

(b) Write down **one** safety risk in Oliver's practical.

Suggest how Oliver controls this risk.

safety risk

.....

how Oliver controls this risk

.....

[2]

(c) After a few minutes the biscuits start to move.

Describe the movement of the biscuits and explain why they move.

You may draw a labelled diagram to help you with your explanation.

description

explanation

.....

.....

.....

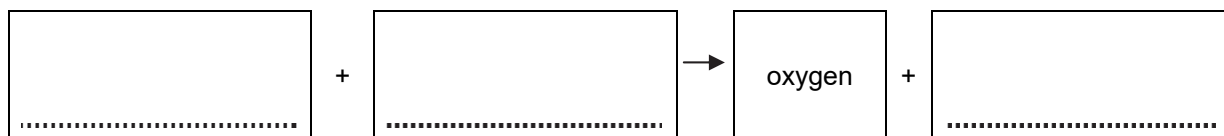
[3]

6 Plants make their own food by photosynthesis.

(a) Name the structure inside a plant cell where photosynthesis takes place.

..... [1]

(b) Complete the **word** equation for photosynthesis.



[2]

(c) Ahmed investigates how light intensity affects the rate of photosynthesis.

Ahmed measures the volume of oxygen produced in 10 minutes by a plant at **increasing** light intensities.

Here are his results.

increasing light intensity	1	2	3	4	5
volume of oxygen produced in 10 minutes in cm ³	3	6	4	11	15
rate of oxygen produced in cm ³ /hour	18		24	66	84

(i) There is **one** anomalous volume of oxygen produced in 10 minutes.

Circle this anomalous **volume** in the table.

[1]

(ii) Ahmed calculates the rate of oxygen produced.

He uses this equation.

rate of oxygen produced = volume of oxygen produced in 10 minutes × 6

Calculate the rate of oxygen produced at light intensity **2**.

Write your answer in the table.

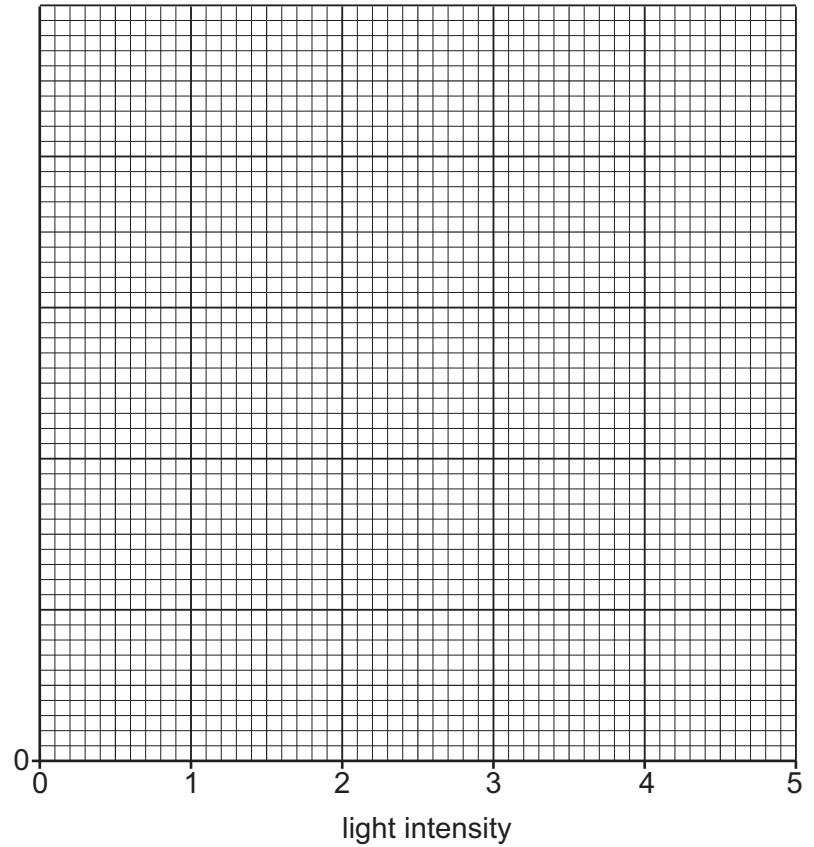
[1]

(iii) Draw the graph of **rate of oxygen produced** against the light intensity.

Include:

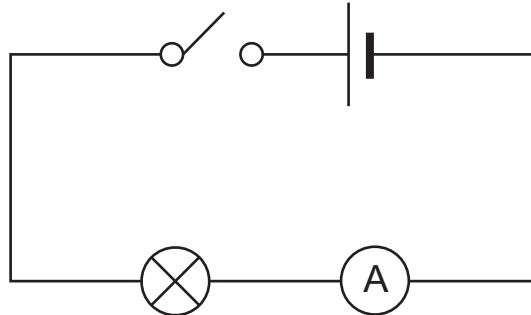
- label for the y-axis including the scale
- plots for **all** the points
- a line of best fit.

.....
.....
.....



[3]

7 Safia makes this electrical circuit.



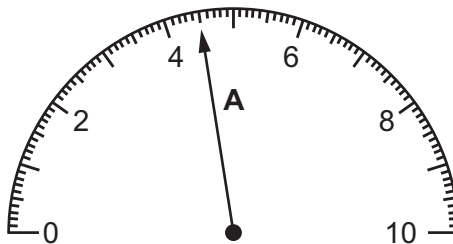
- (a) (i) She measures the voltage across the lamp using a voltmeter.

Complete the circuit diagram to show the correct position of the voltmeter to measure the voltage across the lamp.

Use the correct standard conventional electrical symbol.

[1]

- (ii) Look at the diagram of the ammeter.



What is the reading shown on the ammeter?

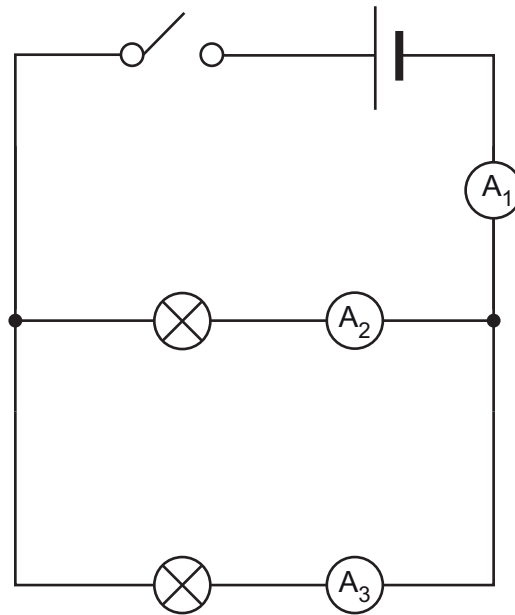
..... A [1]

- (iii) Safia adds an identical lamp in series to her electrical circuit.

Describe the effect of adding this lamp to the electrical circuit.

.....
 [1]

(b) Safia makes this electrical circuit.



Both lamps are identical.

Complete the readings on the ammeters.

$$A_1 = 6 \text{ A}$$

$$A_2 = \dots\dots\dots \text{ A}$$

$$A_3 = \dots\dots\dots \text{ A}$$

[1]

(c) Lily makes a different electrical circuit.

The current through her lamp is 2.5 A.

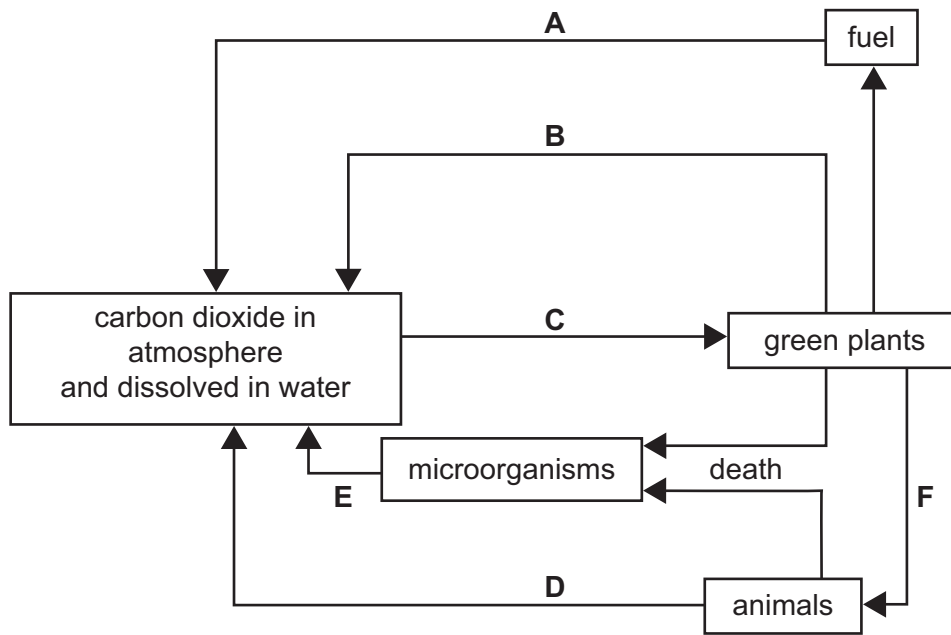
The voltage across her lamp is 12.0 V.

Calculate the resistance of her lamp.

Include the unit of resistance in your answer.

resistance = unit [3]

8 The diagram shows the carbon cycle.



Write down the name of the processes shown by arrows **A** and **F**.

A

F

[2]

9 Tasmanian tigers are extinct.

(a) Describe what is meant by the word **extinct**.

..... [1]

(b) Write down **two** reasons why animals become extinct.

1

.....

2

.....

[2]

10 Look at the table of data about the melting points of different substances.

substance	melting point in °C
graphite	3600
iron	1500
oxygen	−218
water	0

Name **one** substance in the table with a simple structure

.....

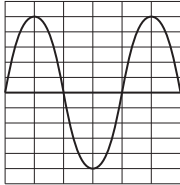
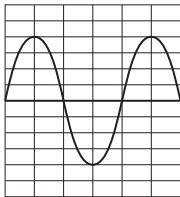
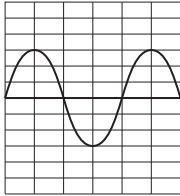
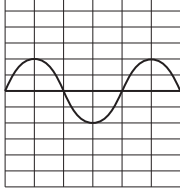
[1]

11 Jamila investigates the loudness of sound at different distances from a sound source.

Jamila:

- switches on the sound source
- measures and records the waveform at different distances from the sound source.

Look at her results.

distance from sound source in m	waveform
0.1	
0.4	
0.7	
1.0	

(a) Describe how the loudness of the sound changes as the distance from the sound source changes.

Explain your answer.

Use information from the waveform in the table.

description

explanation

[2]

(b) Describe how Jamila makes her results more reliable.

.....

[1]

12 Youssef makes the soluble salt zinc chloride.

He uses insoluble zinc carbonate.

(a) He adds an excess of zinc carbonate to an acid.

Name the acid he uses.

..... [1]

(b) When the reaction has finished Youssef has a mixture of insoluble zinc carbonate and zinc chloride solution.

Describe how he separates this mixture to get the zinc chloride solution.

.....
.....
.....
..... [2]

(c) Describe how Youssef makes zinc chloride crystals from the zinc chloride solution.

.....
..... [1]

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The Periodic Table of Elements

Group																		
1	2	1 H hydrogen 1										3	4	5	6	7	8	
		Key atomic number atomic symbol name relative atomic mass																
3 Li lithium 7	4 Be beryllium 9																	
11 Na sodium 23	12 Mg magnesium 24																	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium	88 Ra radium	89–103 actinoids		104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
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actinoids

89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —
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